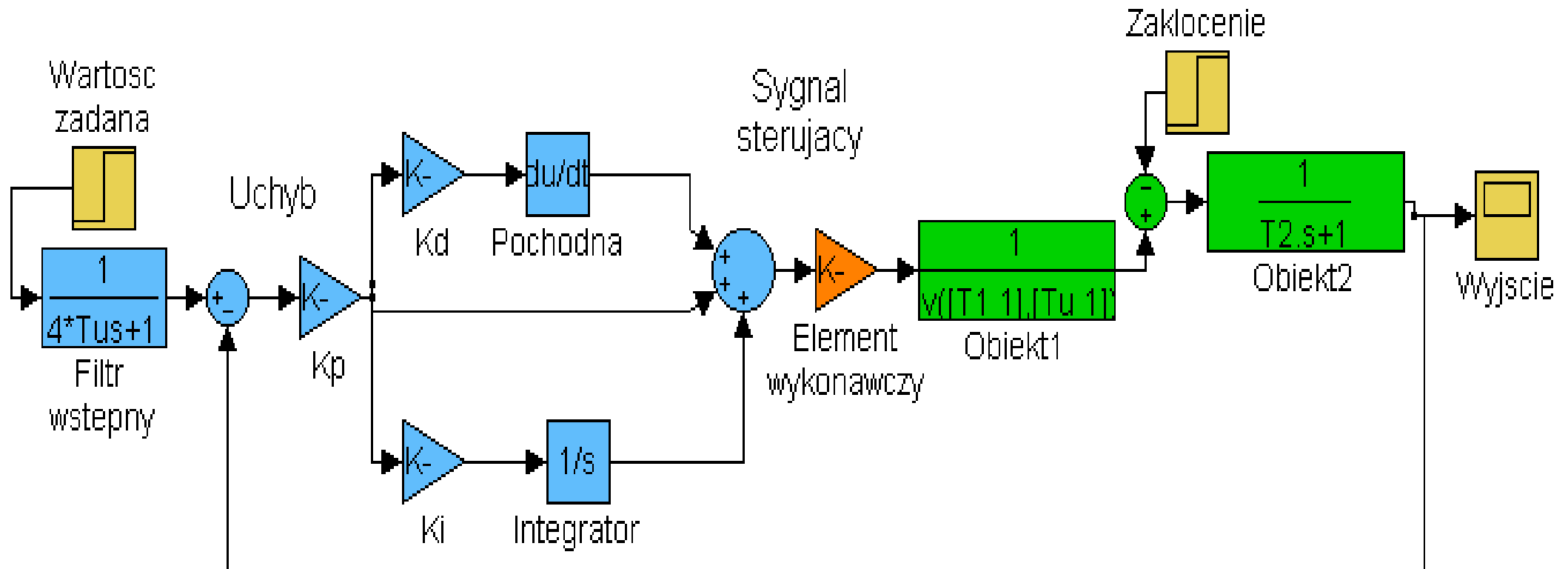


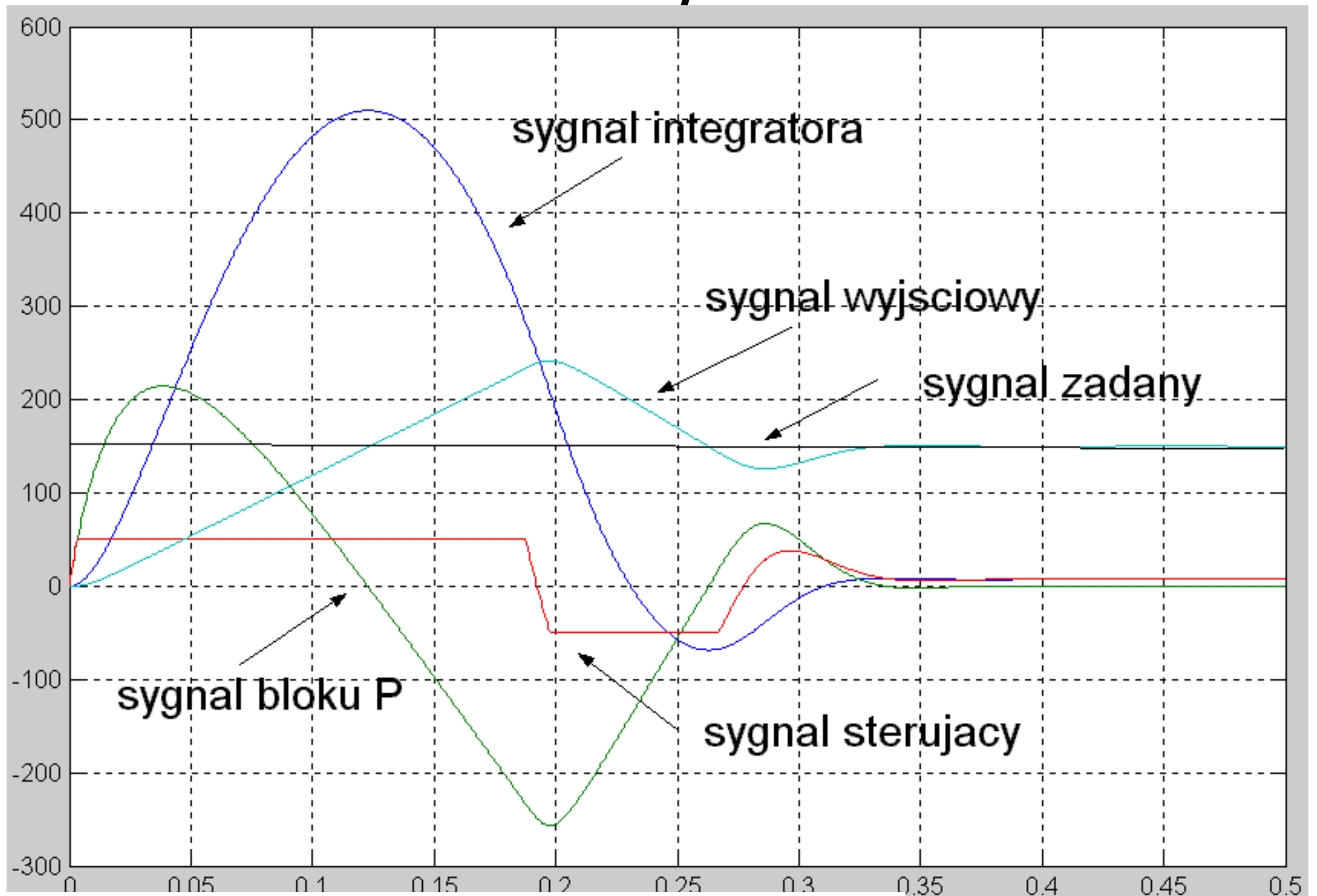
Saturation effect in digital PID algorithm implementation.

PID controller scheme without restrictions

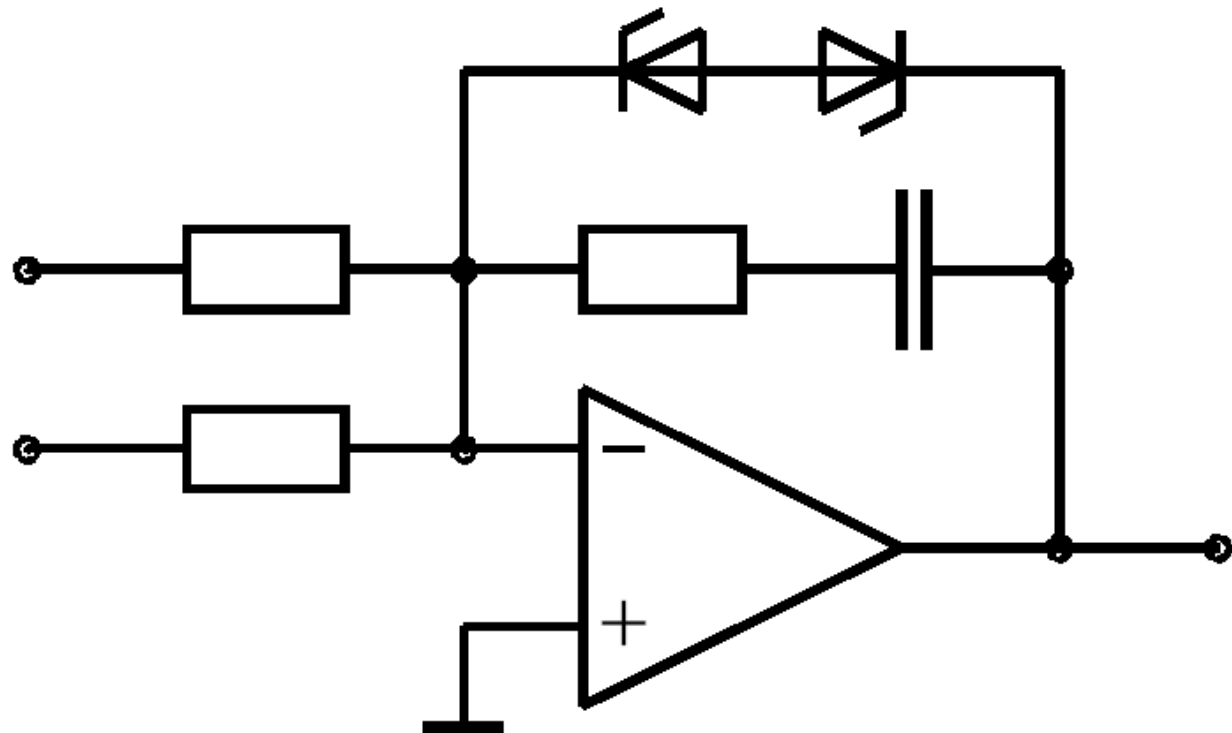


$$u = K_P(e + K_i \int e + K_d(de/dt))$$

Wind-up effect



Elimination of the wind-up effect in the analogue system



Anti-wind-up solutions

- PI controller with variable limits
- Conditional integration
- Limited integration
- Anti-wind-up tracking system
- Modified, tracking system
anti wind-up.
- Error governor

PI controller with variable limits

- The variable limit is introduced in a branch with an integrator so that the total output signal does not exceed the set values.
- Algorithm for PI regulator
- Depending on the fault sign, the member integrating ui in each step is set to the value of respectively **$\min(ui, umax - u_p)$** or **$\max(ui, umax - u_p)$** .

PI controller with variable limits

1. $u_p = K_p e$ and $u_i^* = u_i + K_i \Delta t \cdot e$

2. when $e = 0$ then $u_i = u_i^*$

3. when $e > 0$ to $u_i = \min(u_i^*, u_{\max_up})$

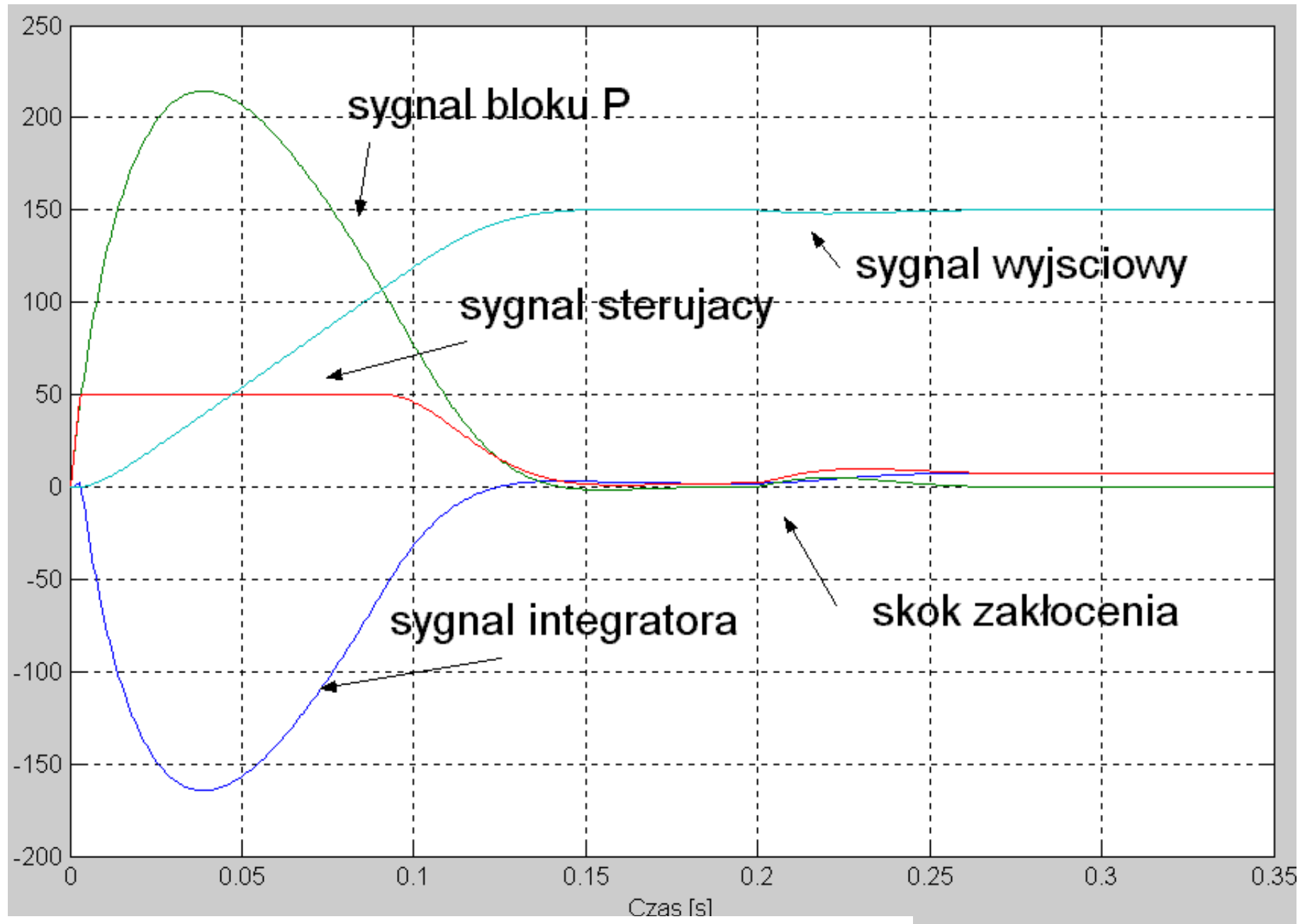
there is no saturation is

saturation

4. when $e < 0$ to $u_i = \max(u_i^*, u_{\min_up})$

5. $u = u_p + u_i$

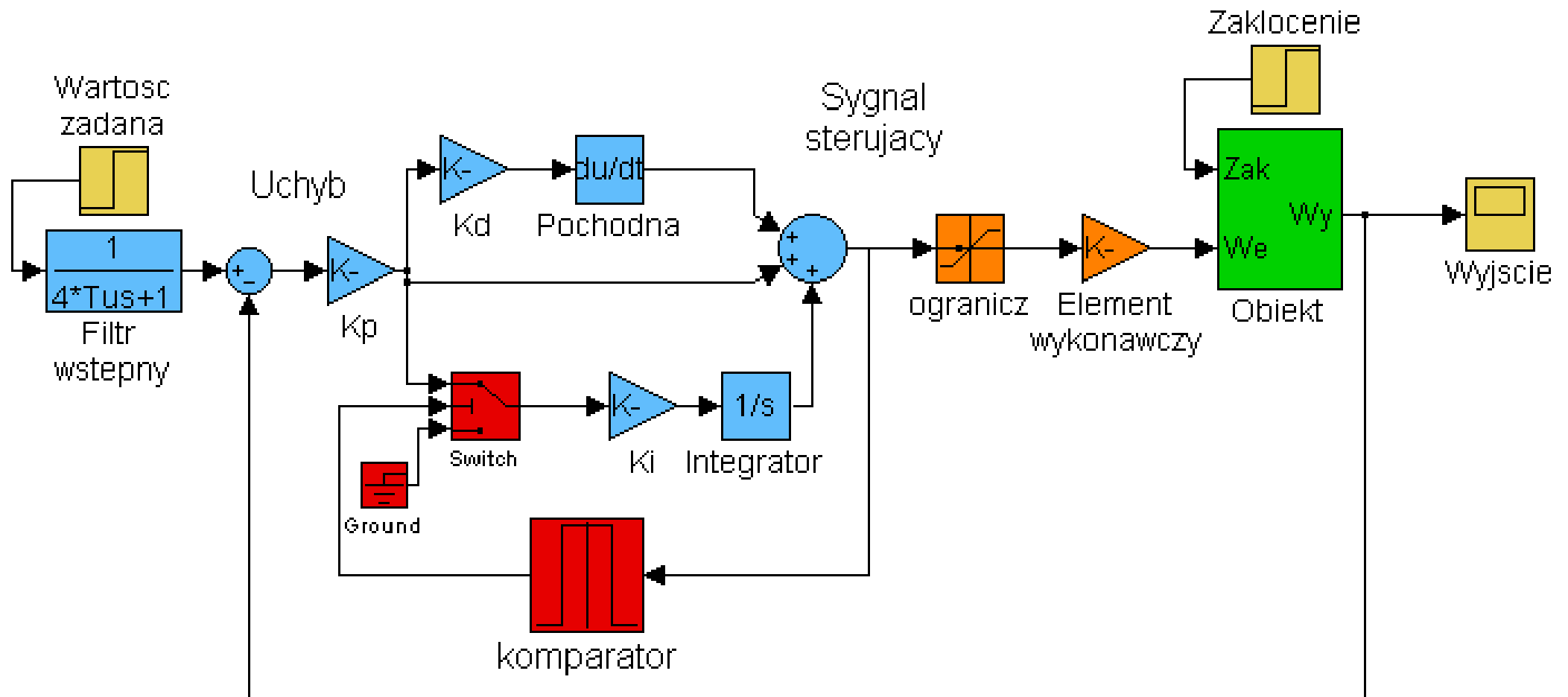
PI controller with variable limits



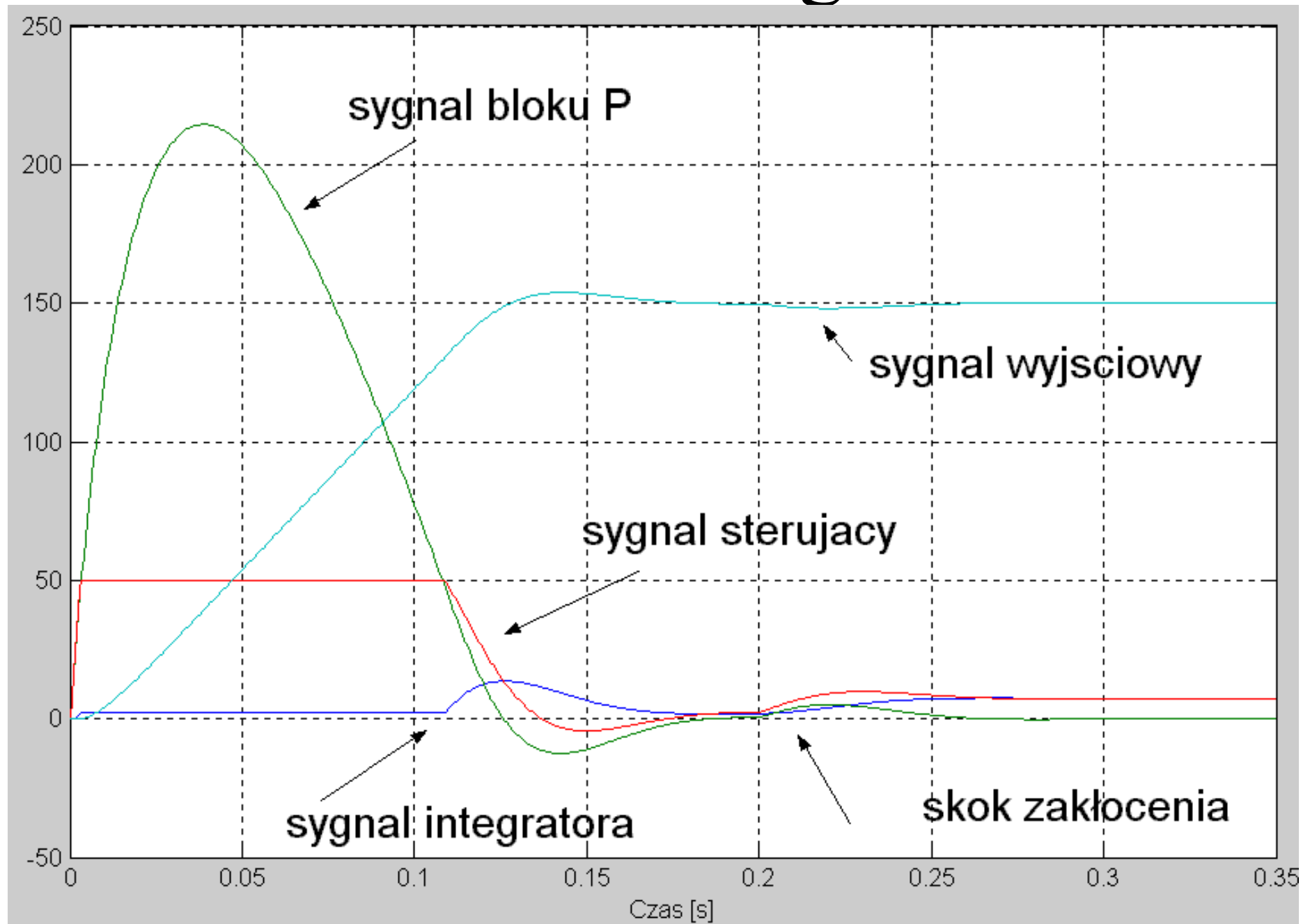
Conditional integration

- In this method, the integrator is switched on or off depending on the conditions:
 - the size of the output signal of the controller
 - of a misconduct signal.
- The best results are achieved when integration is switched off when the output signal is limited and both the output signal and the fault signal have the same character.

Conditional integration



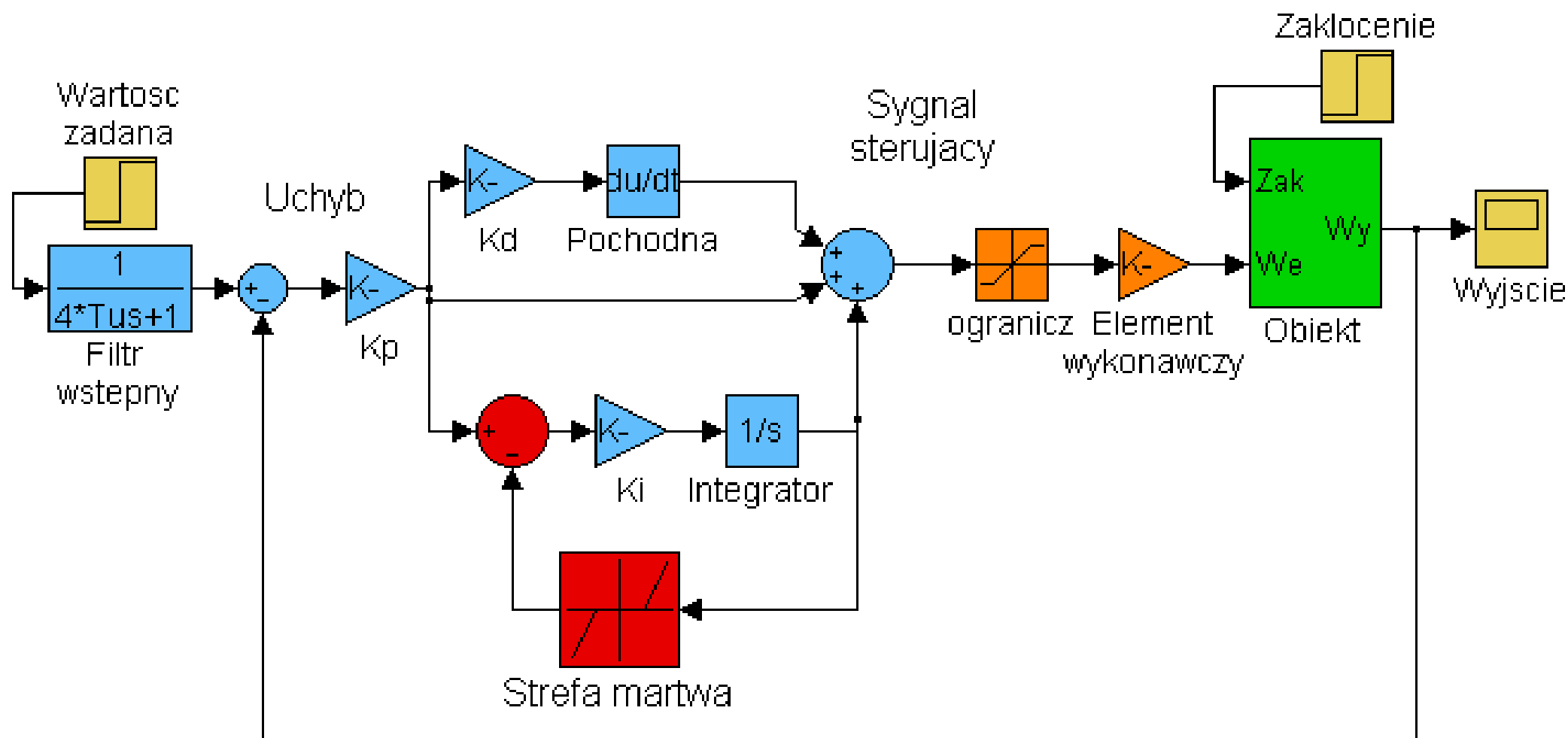
Conditional integration



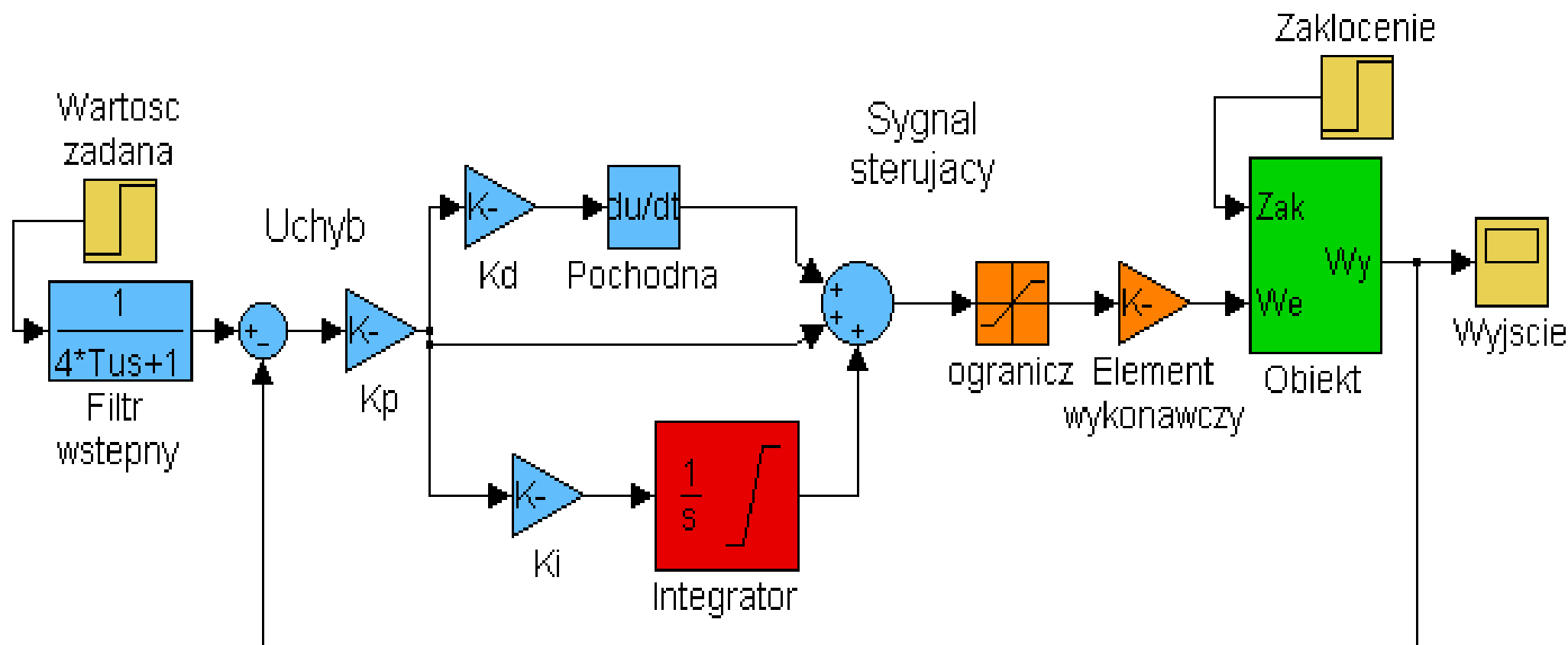
Limited integration

- An additional signal from the integrator output is given to the integrator input, processed by a dead zone block with a high gain factor.
- The size of the dead zone is equal to the range of permissible output signal of the controller.
- In this way, the integrator output signal is limited to the level determined by the dead zone width

Circuit diagram with integrator limitation



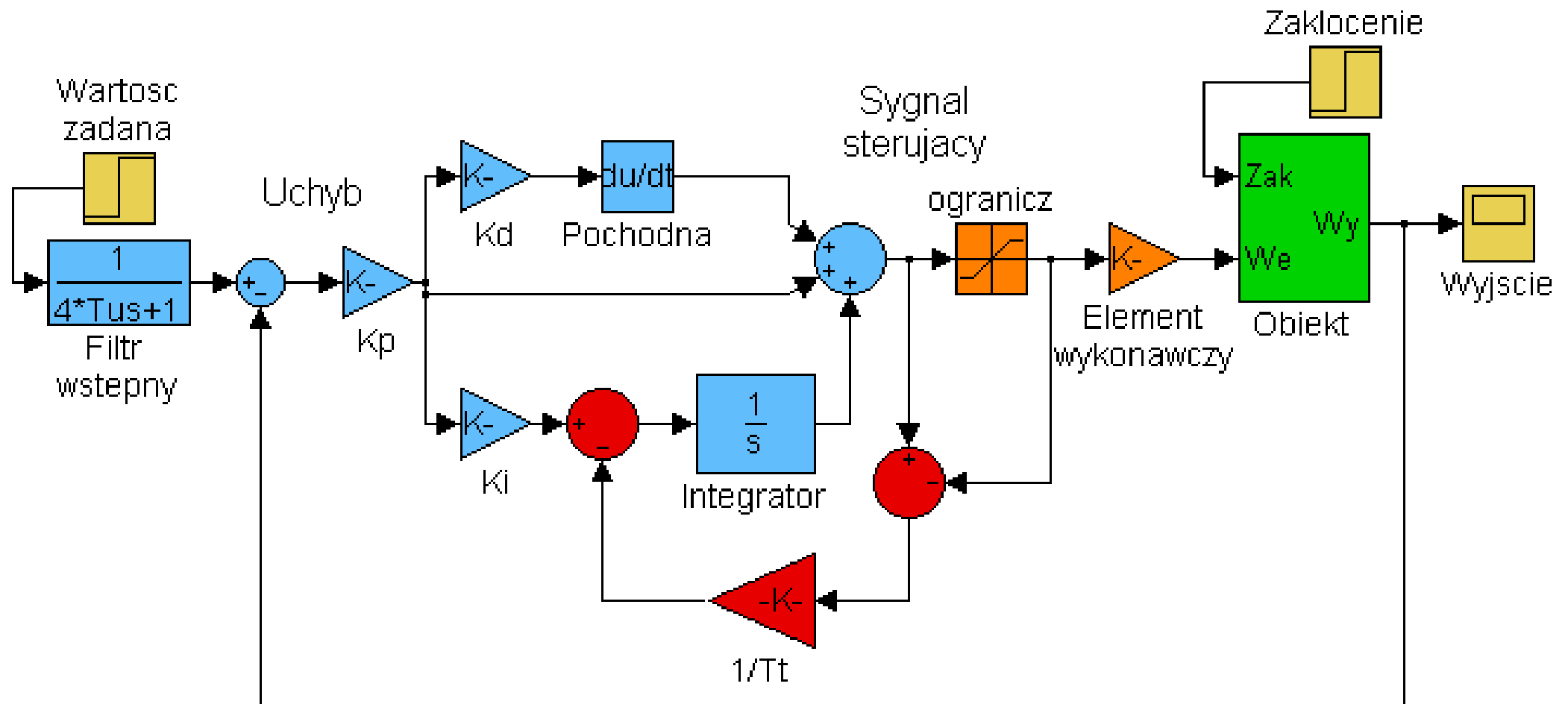
Circuit diagram with integrator limitation



Anti-wind-up tracking system

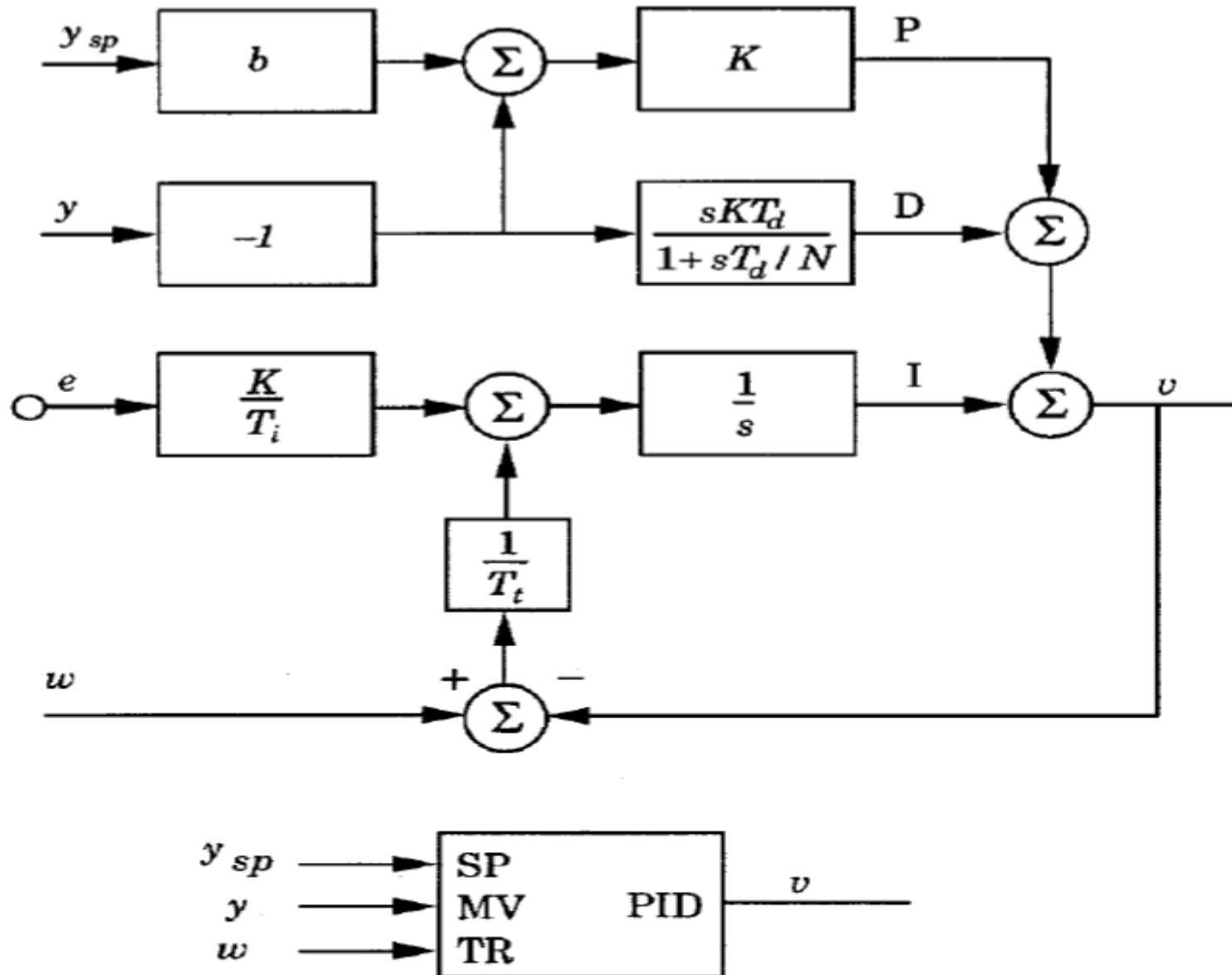
- The classic solution to the *wind-up* problem.
- When the output signal from the controller exceeds the permissible range, a feedback signal is generated.
- This signal is equal to the difference between the signals before and after the restriction.
- The restriction can be actually introduced by the actuator or it can be modelled in the controller.

Anti-wind-up tracking system



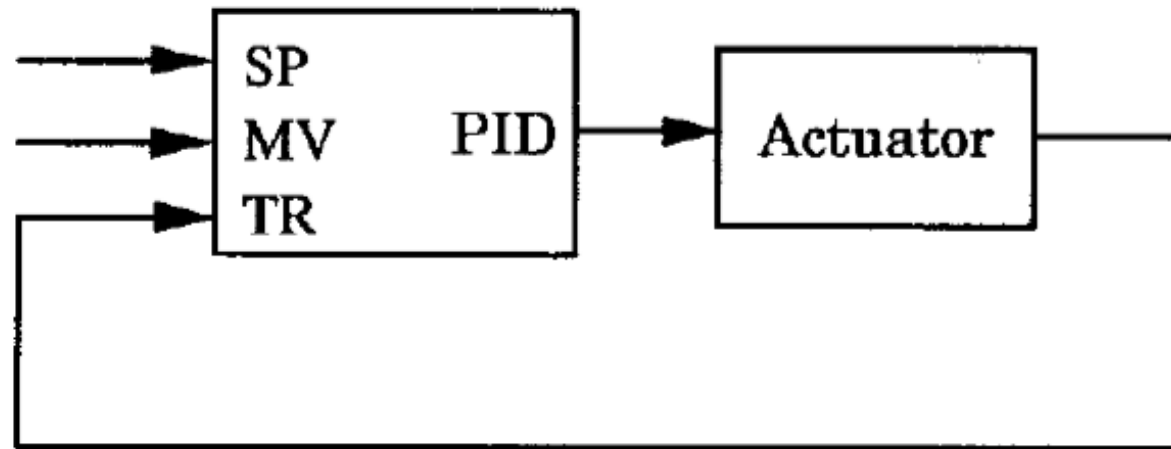
$$k=1/TtTtt=(0.1 \dots 1)* Ti$$

Industrial PID controllers

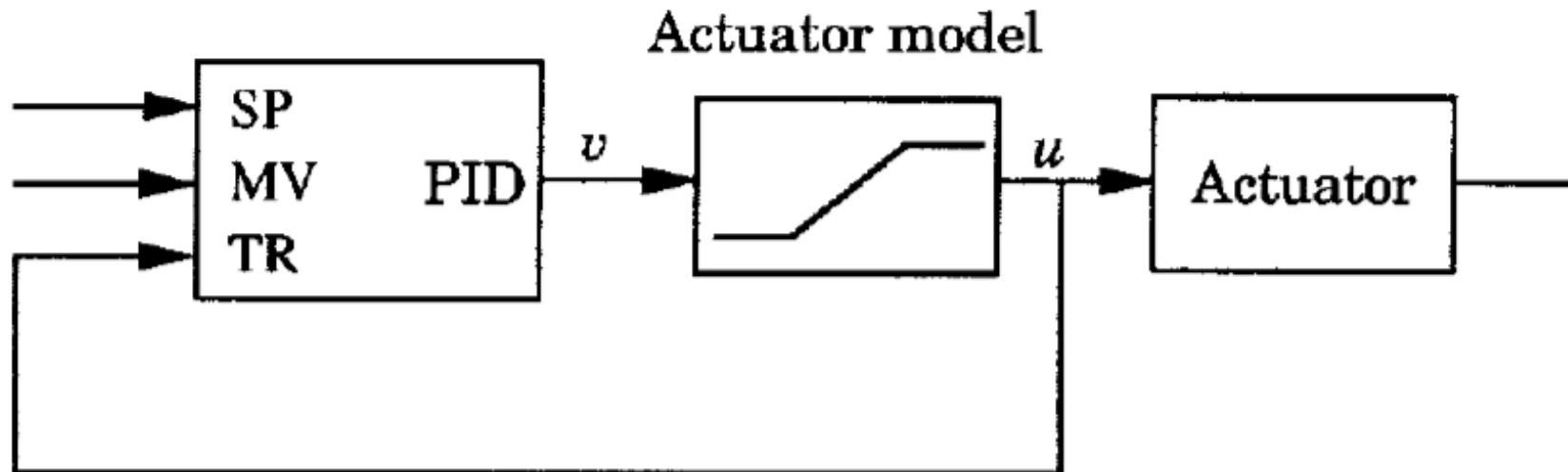


Industrial PID controllers

A



B



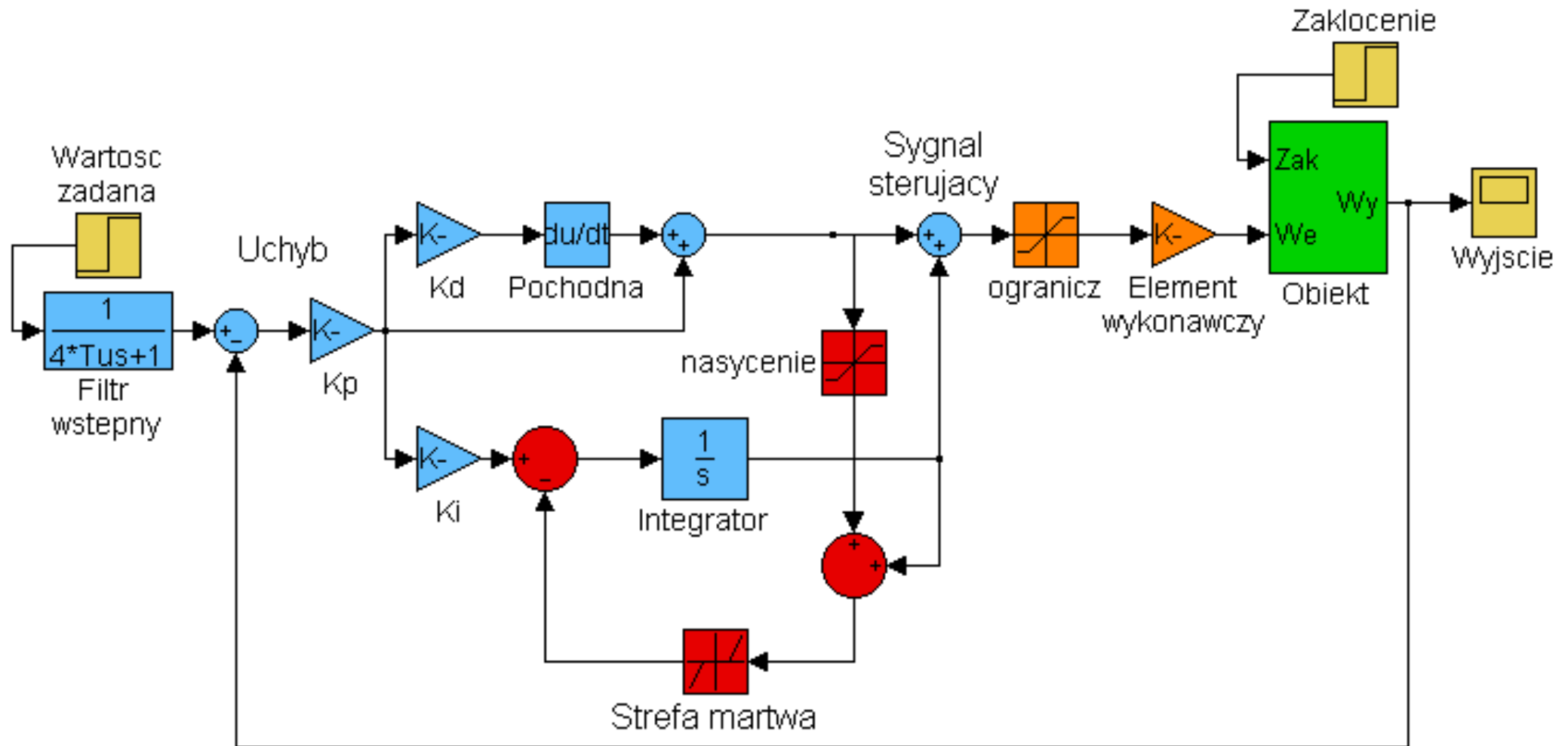
Modified *anti-wind-up* tracking system

- In a classic *anti-wind-up* tracker, a very large initial output signal (resulting from the P and D elements) causes a large feedback signal affecting the integrator.
- Over time, the influence of members P and D decreases, but member I cannot compensate for this change quickly enough.
- The result is a slow response of the regulator.

Modified *anti-wind-up* tracking system

- An additional restriction was introduced only for the PD member.
- This method may be interpreted as halting the integration action until the output of the PD member exceeds the linear range.

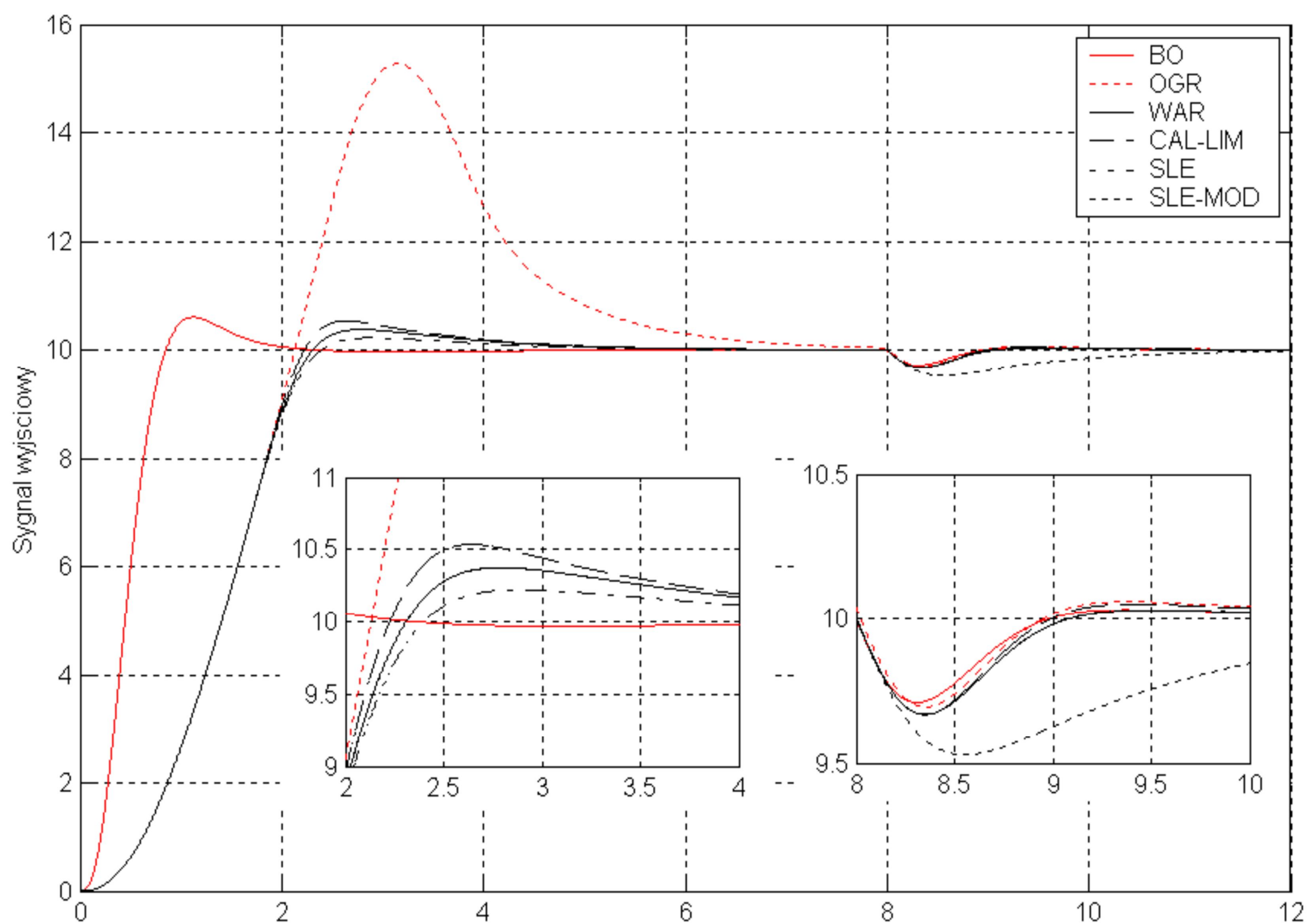
Modified anti-wind-up tracking system

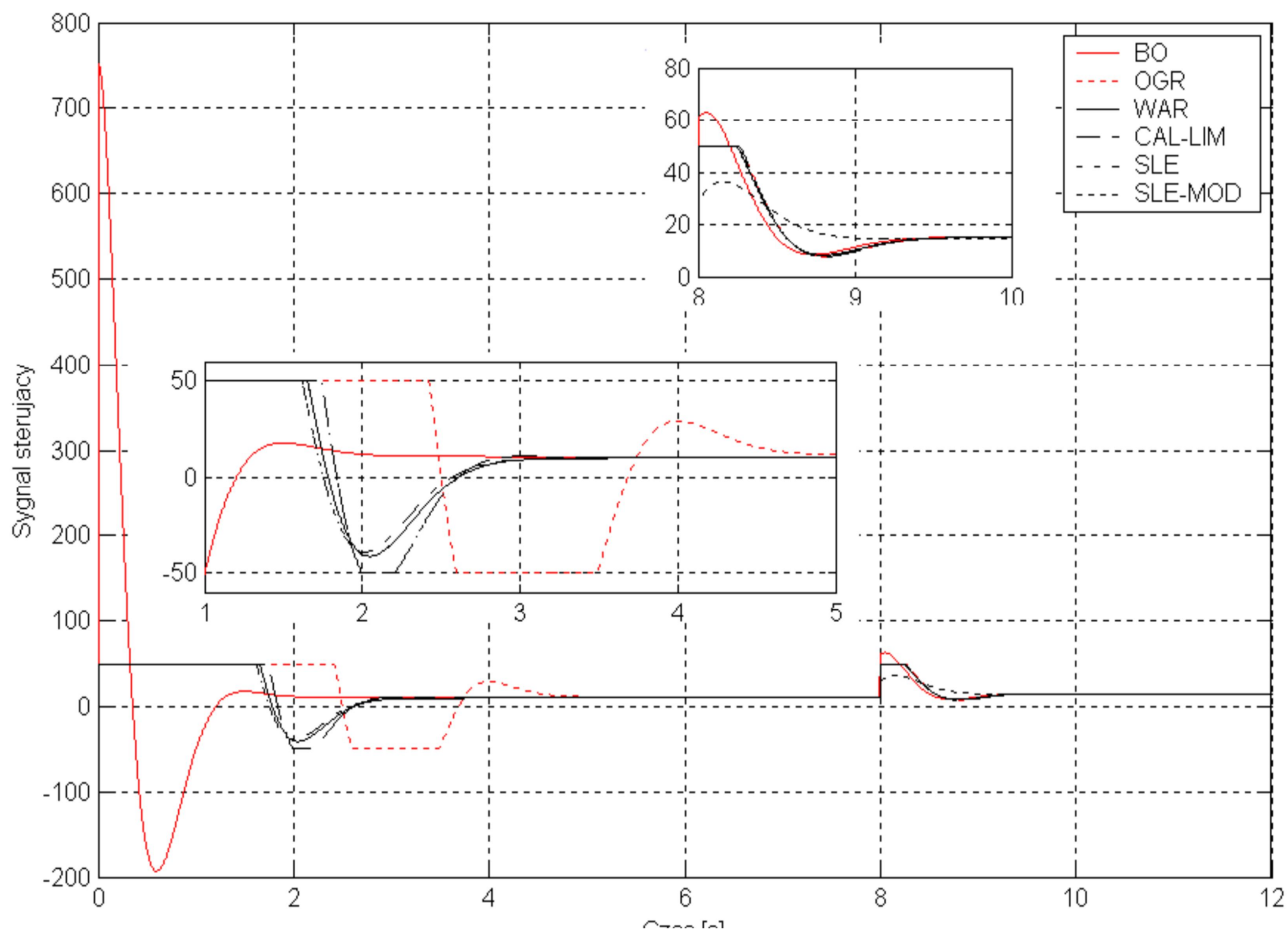


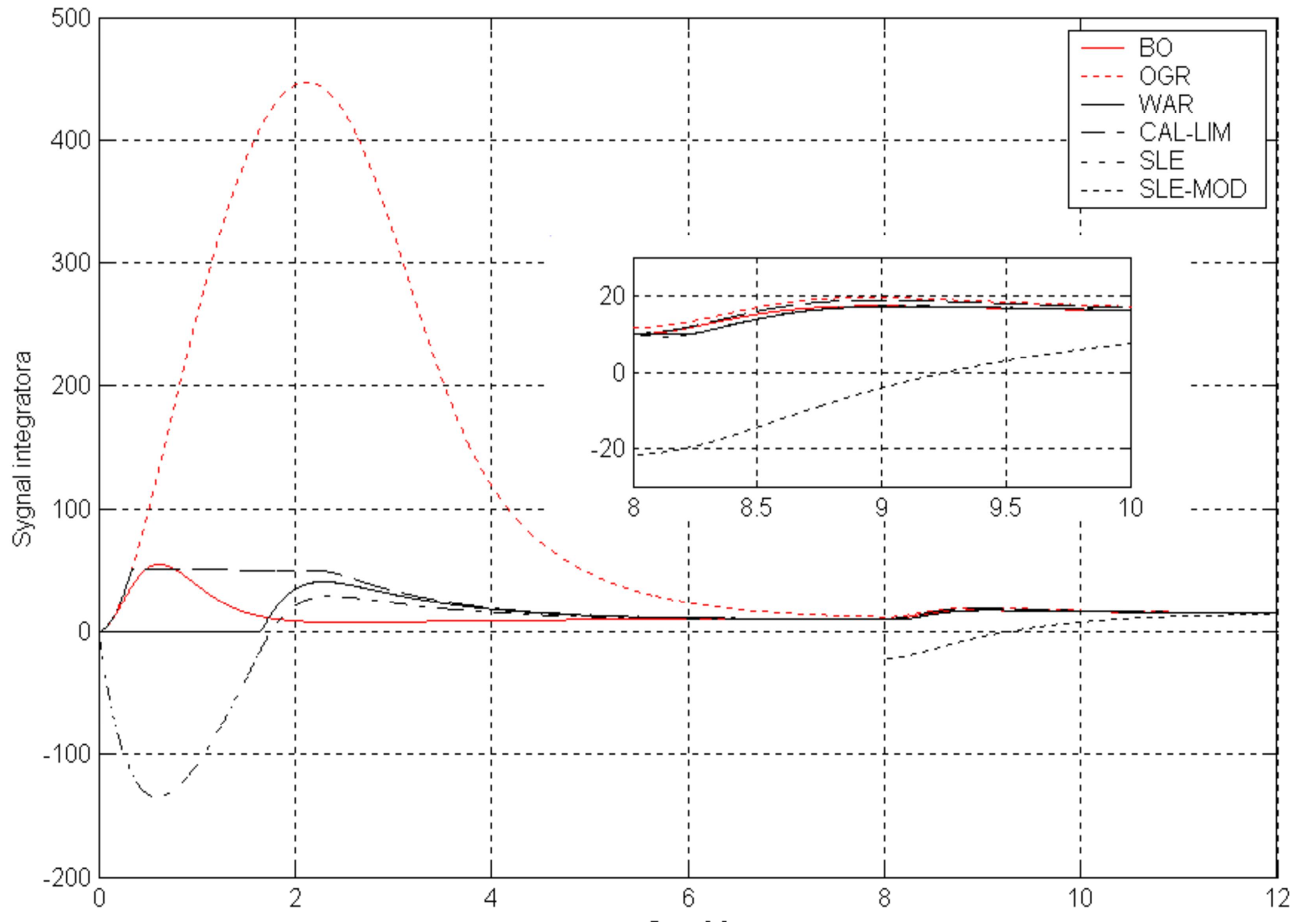
Simulation model data

$$G = \frac{\bullet \text{ Object}}{(1 + sT_1)(1 + sT_2)(1 + s\tau)}$$

- $T_1 = 2 \text{ s}$ $T_2 = 3 \text{ s}$ $\tau = 0.1 \text{ s}$
- **Regulator:**
- $K_p = 75$ $K_d = 0.4$ $K_I = 0.625$
- **Signals:**
- setpoint: jump from 0 to 10 for $t = 0 \text{ s}$
- Interference: Jump from 0 to 5 for $t = 8 \text{ s}$







Error governor

- The presented methods are based on the assumption that the saturation is an effect of the integrating element and therefore they apply correction in the work of the integrator.
- The management method does not require such an assumption.

Management of the infringement

- An idea:
 - Design of the internal control loop without consideration of the saturation phenomenon and then design of the master loop
 - The system operates linearly when the control signal is small (between limits)
 - The control law shall be modified as necessary
- Advantages
 - Limitation of the total output signal
 - For MIMO: to maintain the direction of the control vector.

- Prediction of the control signal in each step.
 - when $|u(t)| < u_{sat}$, then $e(t) = e(t)$
 - when $|u(t)| = u_{sat}$, then $e(t) = e(t)$ so that :

$$\frac{du(e_\lambda)}{dt} = 0$$

- when $|u(t)| > u_{sat}$, then $e(t) = e_{(t)}$ so as to return to u_{sat} .
- Assuming that (for $t = 0$, $e(t)$: $u(e(t_0)) < u_{sat}$
 - the signal u has been shown to remain limited and the system is stable

Use of EG for PID control

$$u = k_d \dot{e} + k_p e + k_i \int e \quad \rightarrow \quad \dot{u} = k_d \ddot{e} + k_p \dot{e} + k_i e = 0$$

Solution: $e(t) = Ae^{1t} + Be^{2t}$

$$\alpha_{1,2} = -\frac{k_p}{2k_d} \pm \sqrt{\left(\frac{k_p}{2k_d}\right)^2 - \frac{k_i}{k_d}}$$

- Coefficients A, B - depend on initial conditions

Summary

- The type of *anti-wind-up* system used has a significant impact on the characteristics of the system (over-regulation, dynamic failure)
- The classic *anti-wind-up* tracker does not always provide optimum performance
- Good results are provided by conditional integration methods and variable constraints.
- An interesting alternative may be the infringement management systems.